

C952

Computer Architecture

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A single-threaded program runs 2 billion instructions with an average CPI of 1.5 on a 3 GHz processor. The chip has 8 cores and 32 MB of L3 cache. What is the CPU time?

- A. 9.0 seconds
- B. 0.67 second
- C. 1.5 seconds
- D. 1.0 second

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A program of 4 billion instructions runs in 1 second on a 2 GHz processor. What is its average CPI?

- A. 0.5
- B. 2 billion
- C. 2
- D. 4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A vendor's marketing sheet promises a \$12,000 accelerator that makes a video filter 4 times faster. The filter takes 40% of total run time. What is the overall speedup?

- A. 4.0
- B. 1.6
- C. About 1.82
- D. About 1.43

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Making part of a program infinitely fast would give a maximum overall speedup of 5. What fraction of run time is that part?

- A.** 100%, since the part becomes infinitely fast
- B.** 80%
- C.** 5%
- D.** 20%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A benchmark runs in 10 seconds on machine A, which costs \$2,400, and in 15 seconds on machine B, which costs \$1,800. Both run the same operating system. How many times faster is A than B?

- A.** 0.67
- B.** 5
- C.** 1.5
- D.** 1.33

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Machine X is 1.25 times faster than machine Y. Y runs a job in 20 seconds. How long does X take?

- A.** 25 seconds
- B.** 15 seconds
- C.** 16 seconds
- D.** 20 seconds

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A 5-stage RISC-V pipeline at 2.4 GHz with full forwarding runs `lw x1, 0(x2)` followed immediately by `add x3, x1, x4`. The code was compiled by a team in Austin. How many stall cycles are needed?

- A. It is a control hazard, so the pipeline flushes.
- B. 0, because forwarding removes all data hazards
- C. 2
- D. 1

Target word	The one fact that decides	Answer	☐ sure ☐ unsure ☐ guess
_____	_____	___	

8.

An 8-way set-associative L1 cache in a 3 GHz processor has a hit time of 1 cycle, a miss rate of 5%, and a miss penalty of 100 cycles. What is the average memory access time?

- A. 96 cycles
- B. 6 cycles
- C. 5 cycles
- D. 101 cycles

Target word	The one fact that decides	Answer	☐ sure ☐ unsure ☐ guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A cache has an AMAT of 3 cycles, a hit time of 1 cycle, and a miss penalty of 50 cycles. What is its miss rate?

- A. 2%
- B. 6%
- C. 4%
- D. 50%

Target word	The one fact that decides	Answer	☐ sure ☐ unsure ☐ guess
_____	_____	___	

10.

A loop reads `a[0]`, `a[1]`, `a[2]`, and so on through an array of 1 million floats collected by a weather station in Kansas. Each cache block holds 16 floats. Which property makes the cache effective here?

- A.** Spatial locality
- B.** Temporal locality
- C.** Caches only help when the same address is reused.
- D.** Virtual memory paging

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A payroll database server for 9,000 employees has a 3.2 GHz CPU. Its disk controller interrupts the CPU for every word of each 4 MB block it transfers, and the CPU spends most of its time handling I/O. What change best reduces this load?

- A.** Use direct memory access (DMA) so the CPU is interrupted only when a block finishes.
- B.** Upgrade to a faster CPU clock.
- C.** Raise interrupt priority so each word is handled sooner.
- D.** Switch to polling the controller.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

In a 4-core server with 64 GB of DDR5 memory, core 1 and core 2 both cache shared variable `x`. Core 1 writes a new value to `x`. What ensures core 2 reads the new value?

- A.** Message passing between the cores
- B.** A cache coherence protocol that invalidates or updates core 2's copy
- C.** A larger L1 cache on core 2
- D.** A write-back policy on core 1's cache

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A single-threaded program runs 2 billion instructions with an average CPI of 1.5 on a 3 GHz processor. ~~The chip has 8 cores and 32 MB of L3 cache.~~ What is the CPU time?

Target word: CPU time **Decides:** $2 \times 10^9 \times 1.5 \div 3 \times 10^9$.

A	Inverted operation	Multiplies by clock rate instead of dividing.
B	Dropped element	Leaves out CPI.
C	Wrong stem number	That is the CPI.
D	Correct	$3 \times 10^9 \text{ cycles} \div 3 \times 10^9 \text{ cycles per second} = 1.0 \text{ s.}$

2. Answer A Concept 1 • transfer

A program of 4 billion instructions runs in 1 second on a 2 GHz processor. What is its average CPI?

Target word: solve for CPI **Decides:** $\text{CPI} = \text{time} \times \text{clock rate} \div \text{instruction count}$.

A	Correct	$1 \times 2 \times 10^9 \div 4 \times 10^9 = 0.5$.
B	Dropped element	Total cycles; not divided by instructions.
C	Inverted operation	Divides instructions by cycles.
D	Wrong stem number	The instruction count in billions.

3. Answer D Concept 2

~~A vendor's marketing sheet promises a \$12,000 accelerator~~ that makes a video filter 4 times faster. The filter takes 40% of total run time. What is the overall speedup?

Target word: overall speedup **Decides:** $1 \div (0.6 + 0.4 \div 4) = 1 \div 0.7$.

A	Wrong stem number	The part speedup, not the overall speedup.
B	Dropped element	Multiplies 0.4×4 and ignores the unimproved time.
C	Inverted operation	Applies the gain to the 60% instead of the 40%.
D	Correct	$1 \div 0.70 \approx 1.43$.

4. Answer B Concept 2 • transfer

Making part of a program infinitely fast would give a maximum overall speedup of 5. What fraction of run time is that part?

Target word: solve for f **Decides:** $1 \div (1 - f) = 5$.

A	Absolute language	At 100% the speedup would be unlimited, not 5.
B	Correct	$1 - f = 0.2$, so $f = 0.8$.
C	Wrong stem number	Reuses the speedup number as a percentage.
D	Inverted operation	That is the unimproved part.

5. Answer C Concept 3

A benchmark runs in 10 seconds on machine A, ~~which costs \$2,400,~~ and in 15 seconds on machine B, ~~which costs \$1,800. Both run the same operating system.~~ How many times faster is A than B?

Target word: performance ratio **Decides:** $15 \div 10$.

A	Inverted operation	Divides A's time by B's.
B	Adjacent category	That is the difference in seconds.
C	Correct	Time B $\div$ time A = 1.5.
D	Camouflage	The price ratio, not performance.

6. Answer C Concept 3 • transfer

Machine X is 1.25 times faster than machine Y. Y runs a job in 20 seconds. How long does X take?

Target word: solve for X's time **Decides:** Time X = time Y $\div$ 1.25.

A	Inverted operation	Multiplies instead of dividing.
B	Adjacent category	Cuts time by 25%, which is not the same as 1.25 times faster.
C	Correct	$20 \div 1.25 = 16$.
D	Wrong stem number	Y's time.

7. Answer D Concept 4

A 5-stage RISC-V pipeline ~~at 2.4 GHz~~ with full forwarding runs lw x1, 0(x2) followed immediately by add x3, x1, x4. ~~The code was compiled by a team in Austin.~~ How many stall cycles are needed?

Target word: stalls for load-use **Decides:** Load data is ready only after MEM; the next EX needs it one cycle earlier.

A	Confusion set	No branch is involved; this is a data hazard.
B	Absolute language	Forwarding cannot send data back in time from MEM.
C	Dropped element	That is the no-forwarding case.
D	Correct	A load-use hazard needs one bubble even with forwarding.

8. Answer B Concept 5

~~An 8-way set-associative L1 cache in a 3-GHz processor~~ has a hit time of 1 cycle, a miss rate of 5%, and a miss penalty of 100 cycles. What is the average memory access time?

Target word: AMAT in cycles **Decides:** $1 + 0.05 \times 100$.

A	Inverted operation	Uses the 95% hit rate as the miss rate.
B	Correct	$1 + 5 = 6$.
C	Dropped element	Leaves out the hit time.
D	Confusion set	Treats every access as a miss.

9. Answer C Concept 5 · transfer

A cache has an AMAT of 3 cycles, a hit time of 1 cycle, and a miss penalty of 50 cycles. What is its miss rate?

Target word: solve for miss rate **Decides:** $(3 - 1) \div 50$.

A	Confusion set	$1 \div 50$ uses the hit time instead of AMAT – hit time.
B	Dropped element	$3 \div 50$ leaves out the hit time.
C	Correct	$2 \div 50 = 0.04$.
D	Wrong stem number	The miss penalty number.

10. Answer A Concept 6

A loop reads $a[0]$, $a[1]$, $a[2]$, and so on through an array of 1 million floats ~~collected by a weather station in Kansas~~. Each cache block holds 16 floats. Which property makes the cache effective here?

Target word: type of locality **Decides:** Consecutive addresses in the same block.

A	Correct	Loading one block brings in the next 15 neighbors.
B	Confusion set	Each element is read once, not reused.
C	Absolute language	Block fetches also help with nearby addresses.
D	Adjacent category	Paging maps memory; it does not explain the cache hits.

11. Answer A Concept 7

A payroll database server ~~for 9,000 employees has a 3.2 GHz CPU~~. Its disk controller interrupts the CPU for every word of each 4 MB block it transfers, and the CPU spends most of its time handling I/O. What change best reduces this load?

Target word: best I/O method **Decides:** Large block transfers with per-word CPU work.

A	Correct	DMA moves the block without CPU per-word handling.
B	Camouflage	Answers the clock detail; the overhead per word stays.
C	Dropped element	Still one interrupt per word.
D	Adjacent category	Polling keeps the CPU busy checking the device.

12. Answer B Concept 8

In a 4-core server ~~with 64 GB of DDR5 memory~~, core 1 and core 2 both cache shared variable x . Core 1 writes a new value to x . What ensures core 2 reads the new value?

Target word: mechanism for current values **Decides:** Stale copies in private caches.

A	Confusion set	That is a distributed-memory model, not shared-memory hardware.
B	Correct	Snooping or directory protocols such as MESI keep copies consistent.
C	Adjacent category	Size does not fix stale data.
D	Dropped element	Write-back delays the memory update and does nothing to core 2's copy.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	8	
Wrong stem number	A figure copied from the prompt that is not the answer.	6	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	CPU performance equation	CPU time = instruction count $\times$ CPI $\div$ clock rate.	Multiplying by clock rate, or leaving out CPI.	Instruction count, CPI, and GHz in the same stem.
2	Amdahl's law	Overall speedup = $1 \div ((1 - f) + f \div s)$; the unimproved part caps the gain.	Assuming the whole program speeds up by s .	"X% of run time is made N times faster."
3	Comparing performance	Performance = $1 \div$ execution time; "A is n times faster than B" means $n = \text{time B} \div \text{time A}$.	Dividing the faster time by the slower, or using the difference.	Two run times and "how many times faster."
4	Pipeline data hazards	Forwarding removes most ALU-to-ALU stalls, but a load followed immediately by a use still needs one stall in a 5-stage pipeline.	Believing forwarding removes every stall.	A load instruction whose result is used by the very next instruction.
5	Average memory access time	AMAT = hit time + miss rate $\times$ miss penalty.	Leaving out the hit time or using the hit rate as the miss rate.	Hit time, miss rate, and miss penalty given in cycles.
6	Locality of reference	Temporal locality = the same address reused soon; spatial locality = nearby addresses used soon.	Calling sequential array access temporal locality.	A loop walking through consecutive elements, or a variable reused in a loop.
7	I/O methods: polling, interrupts, DMA	DMA moves blocks between a device and memory without the CPU handling each word; the CPU is interrupted only at completion.	Thinking a faster CPU or polling fixes per-word I/O overhead.	Large block transfers with high CPU load from I/O.
8	Cache coherence in multiprocessors	In shared-memory multicore systems, a coherence protocol (such as snooping with MESI) invalidates or updates stale cached copies after a write.	Assuming each core automatically sees the latest value.	One core writes a shared variable another core has cached.

Course scope (public description): Computer Architecture introduces students to concepts and characteristics of organization and architecture applied to modern computer systems, including performance, processor, memory, I/O, and multiprocessors to optimize system design, performance, and efficiency.